

DUSTBADGE · REQUIREMENTS

DustBadge requirements

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DOCUMENT CONTROL

Document ID	DBG-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dustbadge		
Source	docs/03-requirements.md @ 5045449		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3, status from DBG-CAL-001 for every requirement; R8 restated for the inlet; R6 defaults follow DBG-DDR-001 D3; R11 and R3 now not met, R10 not verifiable at TRL 3
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

DustBadge requirements

These are the requirements for the concept, with their status from the TRL 3 calculations in DBG-CAL-001. On paper four are not met (R2 range, R3 accuracy, R11 full sun at 45 °C and R14 intrinsic safety), one is at risk (R7 run time), one cannot be verified at TRL 3 (R10), and nine are met, five by calculation and four by design. No target was relaxed or redefined by the TRL 2 review decisions (DBG-DDR-001); options for R3 and R7 are proposed in docs/REVIEW.md , awaiting Amish.

Table 1. Requirements and concept status.

ID	Requirement	Target	Status at TRL 3 (DBG-CAL-001)	Verification (TRL 3 or later)
R1	Measure the respirable dust fraction	Mass concentration of particles up to 4 µm (PM4), a proxy for the ISO 7708 respirable convention	Met by design. PM4 is a proxy: for quartz, optical PM4 would include grains up to about 6.5 µm aerodynamic, while the sensor under-sees coarse grains (CAL-001, D)	Datasheet review; later comparison with a cyclone sampler
R2	Cover the working range	0 to 5 mg/m ³ respirable dust without saturating	Not met. 0 to 1,000 µg/m ³ ; the RCS limit lies in range only for silica fractions of 5 % or more; over-range minutes are flagged and counted at the range limit (CAL-001, E)	Calculation of expected site ranges; chamber test on CalRig
R3	Accuracy after site calibration	Shift average within ±25 % of a co-located gravimetric sample	Not met on paper. About ±34 % for quartz-rich stone at the action level (8 h filter collects 0.037 mg), ±21 to ±29 % for a quarry; ±22 % with a 4.2 L/min, two-shift reference (CAL-001, F). Error terms assumed	Error budget at TRL 3; later co-located field samples
R4	Estimate silica exposure	RCS estimate = calibrated PM4 mass x site silica fraction, always labeled as an estimate; dust shown alone when no fraction is entered	Met by design	Firmware sketch review
R5	Update often enough to catch tasks	1 s readings, 1 min logged averages, running 8 h time-weighted average (TWA) updated every minute	Met by design; the sensor gives new readings every second	Firmware sketch review
R6	Warn before a limit is reached	Vibration and LED when the projected 8 h RCS TWA exceeds the action level (default 25 µg/m ³) and a repeated alert at the limit (default 50 µg/m ³); both configurable	Met by design and on paper: five scenarios alert as intended, one early warning after a single heavy cut (CAL-001, G). Defaults per DBG-DDR-001 D3	Firmware sketch review
R7	Last a full shift	12 h of continuous sampling per charge	At risk. 13.3 h at the typical 55 mA and 25 °C; 11.3 h at the 65 mA maximum or at 0 °C (CAL-001, A)	Power budget calculation
R8	Wearable in the breathing zone	Inlet worn within 30 cm of the nose and mouth; clip or harness loop	Met on paper: inlet 242 mm from the nose and mouth at a collar or upper-strap mount; a mount more than 269 mm below them fails (CAL-001, I)	Massing model
R9	Light and small	Mass 120 g or less; no larger than 75 x 55 x 35 mm	Met on paper: 111.8 g; 64 x 52 x 33 mm with the clip (CAL-001, J)	Massing model, then weighing
R10	Survive the site	Electronics splash and dust protected (IP54 target except the sensor air path); inlet facing down;	Not verifiable at TRL 3: 750 to 1,500 g deceleration on a 1.5 m drop; gasketed joint untested (CAL-001, K)	Design review; later drop and spray tests

ID	Requirement	Target	Status at TRL 3 (DBG-CAL-001)	Verification (TRL 3 or later)
		survives a 1.5 m drop onto concrete		
R11	Work in site conditions	0 to 45 °C, 10 to 90 % RH non-condensing; readings flagged when humidity or spray may bias them	Not met in full sun above about 42 °C ambient: shell about 63 °C at 45 °C against the sensor's 60 °C limit; sensor best performance only to 40 °C and 80 % RH; humidity flag now possible with the added RH sensor (CAL-001, C)	Literature and datasheet review
R12	Keep a shift log	At least 30 shifts of 1 min records on the badge; export over Bluetooth Low Energy to the worker's phone	Met on paper: 11.5 kB per 12 h shift, about 182 shifts in 2 MB (CAL-001, H)	Storage calculation
R13	Protect the worker's data	Stored on the badge and the worker's phone; shared with an employer only by the worker's choice	Met by design (DBG-DDR-001 D5)	Design review
R14	Safe in hazardous atmospheres	Certified intrinsically safe for gassy mines and explosive atmospheres	Not met and out of scope for this prototype; it must be labeled not for such use	Design review
R15	Low cost and buildable	Parts \$90 or less; no custom PCB required for the first build	Met on paper: \$88.00 against \$90 (CAL-001, L)	Priced BOM

■ Assumptions

- The respirable convention in ISO 7708 has a 50 % cut near 4 µm aerodynamic diameter; PM4 from an optical sensor is used as a proxy.
- A site silica fraction (the quartz share of respirable dust) is available for each site and task from at least one filter sample analyzed by a laboratory or a field infrared method.
- Default limits follow US OSHA and MSHA (50 µg/m³ limit and 25 µg/m³ action level, 8 h TWA), adopted for TRL 3 pending Amish's review (DBG-DDR-001, D3). Other jurisdictions can be configured, for example the EU binding limit of 100 µg/m³.
- The calculation assumptions (sensor figures, cell, error terms, thermal coefficients) are listed in DBG-CAL-001, Table 1.
- Shifts are up to 12 h; the projected TWA is normalized to 8 h as the regulations define it.